Author Response

Reviewer 1

1. Qianqian Chen and al. report an interresting meta-analsysis about prevalence of lung cancer in CPFE patients showing that lung cancer prevalence is increased in CPFE patients compared to IPF patients.

Response: Thank you for your help and interest in our study.

2. Though the guidelines of IPF have been updated in 2018, it makes sense to keep those of 2011 into account.

Response: Thank you very much for your insightful comments. As you have pointed out, the official ATS/ERS/JRS/ALAT Clinical Practice Guideline for idiopathic pulmonary fibrosis (IPF) was revised in 2018. However, all the studies we included in our review applied the 2011 ATS/ERS/JRS/ALAT guidelines to define IPF and seven of the nine studies were published before the updated guidelines were available in September, 2018. The remaining two studies were published in early 2019.

3. The definition of CPFE could be spelled

Response: We appreciate your helpful suggestion regarding improved readability. We have rewritten the sentences referring to definition of CPFE and included a detailed description in the Methods section of our manuscript. (Page 5, Line 89-94, Highlighted Part)

4. Do the papers report the PFTs results? It would be interesting to analyze the impact of FEV1 and FVC.

Response: We have gladly revised our manuscript according to your constructive comments. A total of six studies described the results of pulmonary function testing (PFT) for patients with CPFE/UIP and IPF alone. We have added the descriptions of PFT parameters to the Results section of our manuscript (Page 8, Line 156-164, Highlighted Part). In addition, an overview of the five PFT parameters are provided in revised Table 3 (Page 10, Highlighted Part) that were involved in more than 3 studies we selected in our manuscript, and these include FVC (% predicted), FEV1 (% predicted), FEV1/FVC ratio, DLCO (% predicted) and DLCO/VA (% predicted). In addition, we found that a significantly lower FEV1/ FVC ratio and DLCO (% predicted) existed in patients with CPFE from five of the six studies that reported PFT results in their studies. Thank you for your insightful comments.

5. Are the ethnicity available?

Response: We are grateful to you for pointing this issue out, which will greatly enhance the quality of this manuscript. We have reviewed the raw data of each study we included in our manuscript and found that the ethnicity of subjects from different studies was unclear although all nine studies included in our meta-analysis were conducted in six different countries. As a result, to avoid any misunderstanding regarding the different locations where the nine studies were conducted, we have replaced the word "population" with " countries where the nine included studies conducted" and revised the relevant sentences of the Result section in our manuscript (Page 11, Line 191-194, Highlighted Part). In addition, we have changed the information mentioned in the figure legend of Figure 4 accordingly.

Reviewer 2

1. thank you for the opportunity to review this manuscrit by Qianqian Chen et al. entitled "Increased Risk of Lung Cancer in Combined Pulmonary Fibrosis and Emphysema Compared with Idiopathic Pulmonary Fibrosis: A Systematic Review and Meta-Analysis". This manuscript is a systematic review and meta-analysis focused on the prevalence of lung cancer in patients with CPFE compared to patients with IPF. The authors found that a significantly higher prevalence of lung cancer in patients with CPFE than those with IPF alone.

Response: Thank you for your help and interest in our study.

2. In the method section the inclusion criteria mentioned as "Outcome: risk of developing lung cancer in CPFE" need to be better clarified. It is not clear what the authors wanted to include with this.

Response: Thank you very much for your constructive suggestions. As you suggested, we have revised the inclusion criteria in the Methods section of our manuscript in a concise and easy-to-understand way and clarified the sentences you mentioned above. (Page 4-5, Line 85-89, Highlighted Part).

3. The manuscript need to be revised by a native English speaker since it appears not clearly written and with several misunderstanding or contradictory concepts. In particular much attention must be paid in the method and results section.

Response: Thank you for your helpful comments that made our study more readable! We have used the recommended SAGE Language Services from the submission guidelines for English language editing (Order No. YMMXW_1). The certificate of English editing by SAGE Language Services will be uploaded when we resubmit our manuscript to the submission system. At the same time, we have

revised some of the sentences in the Methods and Results sections according to the reviewers' comments.

4. What did the author want to assert (in the discussion) with "… we only included CPFE patients with pulmonary fibrosis of IPF subtype in our meta-analysis and performed a subgroup analysis for different emphysema thresholds" ? It doesn't sound clear enough and it appears that method are not properly conducted. Indeed they have never mentioned in the selection criteria that they inclued studies according to the presence of a specific radiologic appearance. Did the authors excluded "probable UIP pattern " or alternative o indeterminate pattern at HRCT? If yes why?

Response: Based on your valuable comments, we have revised the Method sections of our manuscript and added specific descriptions of the fibrotic patterns to the selection criteria in our manuscript (Page 4-5, Line 85-89, Highlighted Part). We have also explained why we included CPFE/UIP patients in our meta-analysis in the Introduction section (Page 4, Line 67-71&73-81, Highlighted Part) and Discussion section (Page 13, Line 226-230, Highlighted Part) of our manuscript for better understanding. The definition of IPF and UIP in our manuscript was in accordance with the 2011 official ATS/ERS/JRS/ALAT IPF guidelines. Based on the guidelines, HRCT fibrotic patterns were divided into three subtypes: UIP pattern, Possible UIP pattern, and Inconsistent with UIP Pattern. A UIP pattern and Possible UIP pattern on HRCT were considered as defined UIP pattern for the diagnosis of IPF, if no surgical lung biopsy was performed (Page 5, Line 94-100, Highlighted Part). We believe that your comments have greatly enhanced the quality of our manuscript. Thank you very much.

5. Since diagnostic criteria for defining CPFE is not fully standardized there is a great heterogeneity of patients included in such studies selected for the meta-analysis that may contribute to influence the interpretation of data. In particular the lack of details regarding the extension of emphysema is a great limitation of this attempt to select the studies. Indeed the risk to overestimate the number of patients with CPFE instead of only IPF with some HRCT feature of centrilobular emphysema is very high.

Response: We totally agree with your insightful comments and we have added descriptions of the limitations as you mentioned to the Discussion section of the manuscript (Page 14-15, Line 266-267, Highlighted Part). As you mentioned, the lack of standardized criteria for CPFE including different emphysema thresholds may contribute to undue diagnosis or less sensitivity for CPFE. Thanks a lot for your comment.